

Simple, highly reliable network communication  
system



# ISE5005D Unmanaged Industrial Ethernet Switch

· Public Utility

· Smart Manufacturing

· Smart City

· Smart Energy

# 1. Product Overview

ISE5005D is a product designed specifically for power, transportation, industrial control, and other demanding industrial environments. It integrates features like wide-temperature and wide-voltage design, as well as protected industrial circuits.

## Product Features:

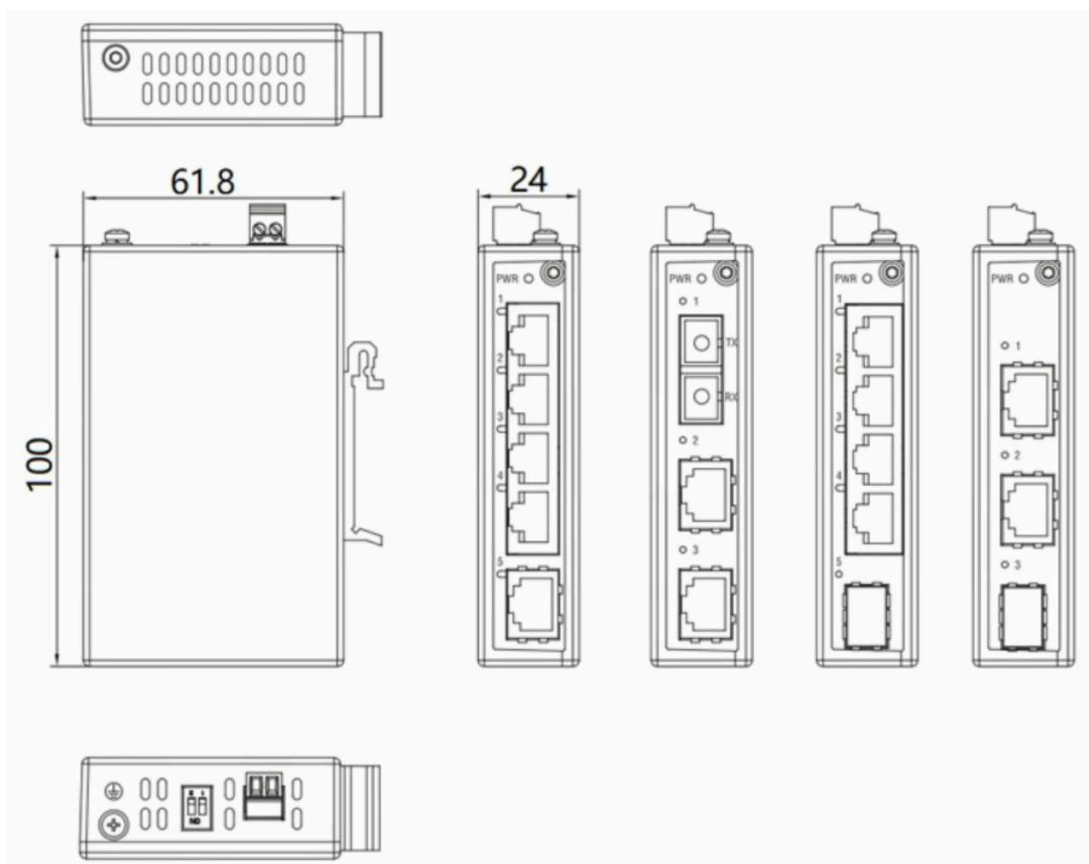
- **Reliability & Resilience:** Fanless design, IP30 protection rating, robust metal casing, and wide-temperature operation.
- **Enhanced Security & Control:** QoS supported to process critical data in heavy traffic.
- **Convenient Deployment:** DIN rail mountable, compact dimensions, and fully plug-and-play.
- **Wide Power Input:** Wide-range power input design.
- **Switching Mechanism:** Possessing a quick store-and-forward mechanism.
- **DIP Switch Integration:** Quality of Service (QoS) and Broadcast Storm Protection (BSP) can be conveniently enabled or disabled through accessible hardware dip switches.
- **International Standard:** Complies with FCC, CE, ROHS, and UL standards.

## Key Technical Specifications

| Parameter             | Specification                                                  |
|-----------------------|----------------------------------------------------------------|
| Type                  | Unmanaged Industrial Ethernet Switch                           |
| Ports                 | 5 x 10/100/1000BaseT (RJ45)                                    |
| Switching Performance | 14 Gbps backplane, Store-and-Forward, 2K MAC                   |
| Dimensions / Weight   | 24 × 100 × 61.8 mm (0.94 × 3.94 × 2.43 in) / 0.23 kg (0.51 lb) |
| Power                 | 9.6-60 VDC & 18-30 VAC, 5 W                                    |

| Parameter      | Specification                                      |
|----------------|----------------------------------------------------|
| Environment    | -40 °C ~ +75 °C (-40 °F ~ +167 °F) operating, IP30 |
| EMC            | IEC 61000-4-2/3/4/5/6/8/18 Class 3+                |
| Certifications | CE, FCC, UL                                        |

## 2. Product Dimensions



ISE5005D

### Note:

1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
2. All dimensions are approximate and **for reference only**.
3. The dimensions shown in the figure **shall not be used for production or processing**.
4. Dimensions must comply with part and manufacturing tolerance requirements.
5. Dimensions are subject to change without notice.

### 3. Hardware Specifications

| Category/Parameter          | Specification                           |
|-----------------------------|-----------------------------------------|
| <b>Physical Performance</b> |                                         |
| Enclosure                   | Fully enclosed seamless metal enclosure |
| Dimensions (W × D × H)      | 24 mm × 100 mm × 61.8 mm                |
| Weight                      | 0.23 kg (0.51 lb)                       |
| Mounting Method             | DIN-rail mounting                       |
| Cooling Method              | Fanless cooling                         |
| Ingress Protection          | IP30                                    |
| Storage Temperature         | -40 °C ~ +85 °C (-40 °F ~ +185 °F)      |
| Operating Temperature       | -40 °C ~ +75 °C (-40 °F ~ +167 °F)      |
| Humidity                    | 5 ~ 95% (non-condensing)                |
| <b>Hardware Performance</b> |                                         |
| Backplane Bandwidth         | 14 Gbps                                 |
| Processing Type             | Store-and-Forward                       |
| MAC Table Size              | 2K                                      |
| Packet Buffer Size          | 1 Mbits                                 |
| Switching Delay             | < 10 μs                                 |

| Category/Parameter                     | Specification                                                                                                                                                                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DIP Switch                             | Quality of Service (QoS), Broadcast Storm Protection (BSP)                                                                                                                                         |
| <b>Power Parameters</b>                |                                                                                                                                                                                                    |
| Operating Voltage                      | 9.6–60 VDC & 18–30 VAC                                                                                                                                                                             |
| Power Consumption                      | 5 W                                                                                                                                                                                                |
| Overcurrent Protection                 | Supported                                                                                                                                                                                          |
| Reverse Polarity Protection            | Supported                                                                                                                                                                                          |
| <b>Electromagnetic Characteristics</b> |                                                                                                                                                                                                    |
| EMS                                    | IEC(EN)61000-4-2, Class 3; IEC(EN)61000-4-3, Class 3<br>IEC(EN)61000-4-4, Class 3; IEC(EN)61000-4-5, Class 3<br>IEC(EN)61000-4-6, Class 3; IEC(EN)61000-4-8, Class 5<br>IEC(EN)61000-4-18, Class 3 |
| <b>Certifications</b>                  |                                                                                                                                                                                                    |
| Certifications                         | CE, FCC, UL                                                                                                                                                                                        |
| <b>Quality Assurance</b>               |                                                                                                                                                                                                    |
| Warranty Period                        | 5 years                                                                                                                                                                                            |
| MTBF                                   | 35 years                                                                                                                                                                                           |

## DIP Switch Settings

Users can use the DIP switch on the device panel to enable/disable Quality of Service (QoS) and Broadcast Storm Protection (BSP) functions. Fast Ethernet ports support 10/100 Mbps auto-negotiation.

| Category/Parameter                     | Specification                                                                                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quality of Service (QoS) – On          | Enables the QoS function, processes packet priorities in 2 WRR queues. Priority 7,6,5,4 → Queue 1 (WRR Weight 16); Priority 3,2,1,0 → Queue 0 (WRR Weight 1) |
| Quality of Service (QoS) – Off         | Disables the QoS function                                                                                                                                    |
| Broadcast Storm Protection (BSP) – On  | Enables broadcast storm protection for all ports (allows max 200 broadcast packets per second)                                                               |
| Broadcast Storm Protection (BSP) – Off | Disables the broadcast storm protection function                                                                                                             |

## 4. Ordering Guide

| Model             | Description                                                                                                                                                                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISE5005D-P-5GT-24 | 5-port Layer 2 unmanaged Industrial Switch. 5 * 10/100/1000BaseT Ports. IP30 Protection Class, Operating Temperature from -40 °C to +75 °C (-40 °F to +167 °F). Single power input: 12/24/48 VDC (9.6~60VDC) or 24 VAC (18~30 VAC) (Only connect to class 2 power supply). |

## 5. Contact Us

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